Year 11 Higher

# REVISION PACK

## Shape and Space 6

**ANSWERS**

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

## Shape and Space 6 · Angles in polygons (Lessons 10.2, 10.3, 10.4)

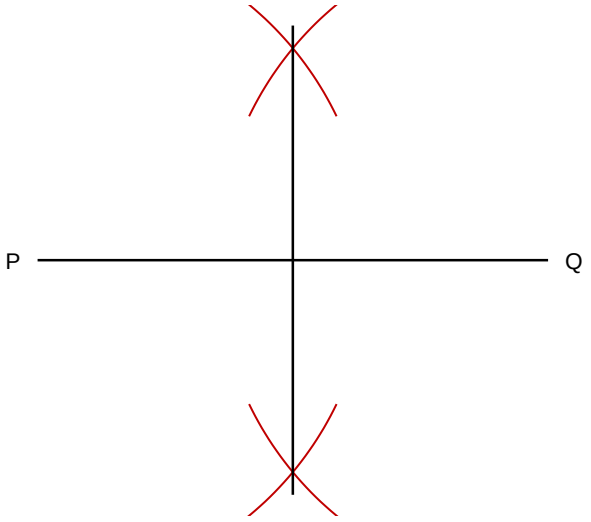
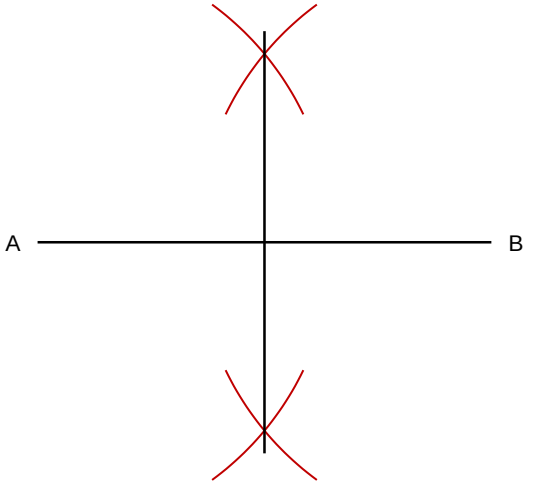
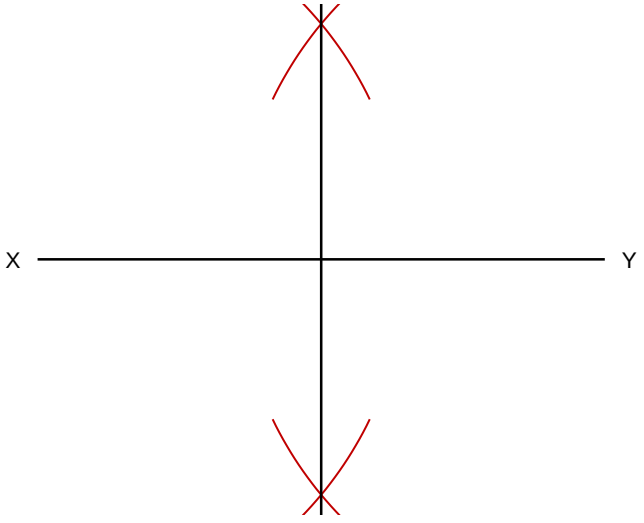
| Q | Working | Answer |
|---|---|---|
| **regular polygons** | | |
| **1.1** | $360 \div 20 = 18$ | **18 sides** |
| **1.2** | $180 - 144 = 36$<br>$360 \div 36 = 10$ | **10 sides** |
| **1.3** | $360 \div 36 = 10$ | **10 sides** |
| **1.4** | $180 - 108 = 72$<br>$360 \div 72 = 5$ | **5 sides** |
| **1.5** | $360 \div 45 = 8$ | **8 sides** |
| **1.6** | $180 - 135 = 45$<br>$360 \div 45 = 8$ | **8 sides** |
| **finding x in a polygon** | | |
| **1.7** | $(6 - 2) \times 180 = 720$<br>$110 + 130 + 140 + 130 = 510$<br>$3x = 720 - 510 = 210$<br>$x = 70$ | $x = 70$ |
| **1.8** | $(5 - 2) \times 180 = 540$<br>$100 + 110 + 120 = 330$<br>$2x = 540 - 330 = 210$<br>$x = 105$ | $x = 105$ |
| **1.9** | $(7 - 2) \times 180 = 900$<br>$140 + 150 + 130 + 120 + 120 = 660$<br>$3x = 900 - 660 = 240$<br>$x = 80$ | $x = 80$ |

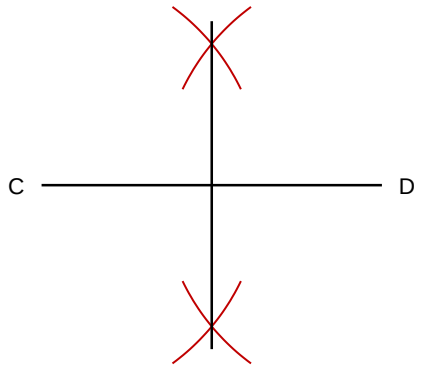
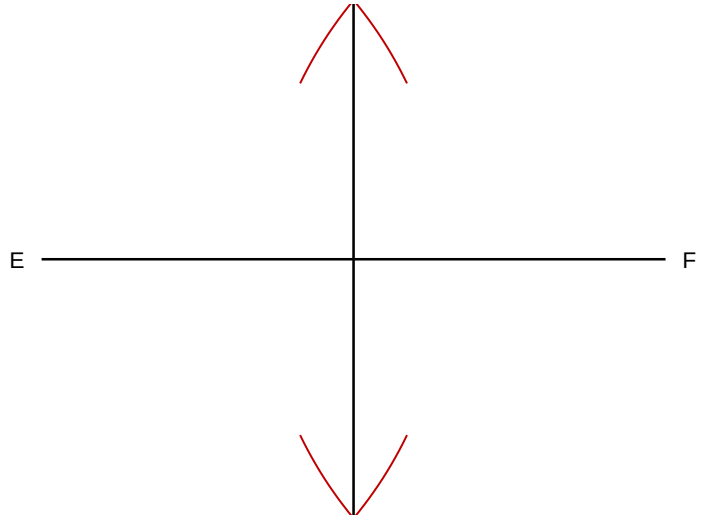
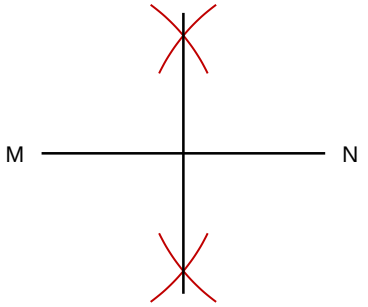
| | | |
|---|---|---|
| **1.10** | $(5-2)\times 180 = 540$<br>$100+140+120 = 360$<br>$2x = 540-360 = 180$<br>$x = 90$ | $x = 90$ |
| **1.11** | $(6-2)\times 180 = 720$<br>$120+125+135+145 = 525$<br>$3x = 720-525 = 195$<br>$x = 65$ | $x = 65$ |
| **1.12** | $(5-2)\times 180 = 540$<br>$90+110+120 = 320$<br>$4x = 540-320 = 220$<br>$x = 55$ | $x = 55$ |
| **angles in a kite** | | |
| **1.13** | $360-64-50 = 246$<br>$246 \div 2 = 123$ | **123°** |
| **1.14** | $360-100-70 = 190$<br>$190 \div 2 = 95$ | **95°** |
| **1.15** | $360-90-40 = 230$<br>$230 \div 2 = 115$ | **115°** |
| **1.16** | $360-56-80 = 224$<br>$224 \div 2 = 112$ | **112°** |
| **1.17** | $360-86-40 = 234$<br>$234 \div 2 = 117$ | **117°** |
| **1.18** | $360-110-34 = 216$<br>$216 \div 2 = 108$ | **108°** |

## Shape and Space 6 · Congruence and similarity (Lessons 10.1, 10.6)

| Q | Working | Answer |
|---|---|---|
| **proving triangles are congruent** | | |

| | | |
|---|---|---|
| **2.1** | $AB = AD$ (given)<br>$CB = CD$ (given)<br>$AC$ is common | **congruent (SSS)** |
| **2.2** | $AB = CD$ (opposite sides)<br>$AD = CB$ (opposite sides)<br>$BD$ is common | **congruent (SSS)** |
| **2.3** | $AB = CD$ (opposite sides)<br>$BC = DA$ (opposite sides)<br>$AC$ is common | **congruent (SSS)** |
| **2.4** | $PX = SX$ (given)<br>$QX = RX$ (given)<br>$\angle PXQ = \angle SXR$ (vertically opposite) | **congruent (SAS)** |
| **similar triangles with parallel lines** | | |
| **2.5** | $(a)\ 2 + 6 = 8$<br>$8 \div 2 = 4$<br>$3 \times 4 = 12$<br>$(b)\ 1.5 \times 4 = 6$<br>$6 - 1.5 = 4.5$ | **(a) 12 cm**<br>**(b) 4.5 cm** |
| **2.6** | $(a)\ 3 + 9 = 12$<br>$12 \div 3 = 4$<br>$4 \times 4 = 16$<br>$(b)\ 2.5 \times 4 = 10$<br>$10 - 2.5 = 7.5$ | **(a) 16 cm**<br>**(b) 7.5 cm** |
| **2.7** | $(a)\ 5 + 3 = 8$<br>$8 \div 5 = 1.6$<br>$7.5 \times 1.6 = 12$<br>$(b)\ 4 \times 1.6 = 6.4$<br>$6.4 - 4 = 2.4$ | **(a) 12 cm**<br>**(b) 2.4 cm** |
| **2.8** | $(a)\ 2 + 3 = 5$<br>$5 \div 2 = 2.5$<br>$4 \times 2.5 = 10$<br>$(b)\ 1.8 \times 2.5 = 4.5$<br>$4.5 - 1.8 = 2.7$ | **(a) 10 cm**<br>**(b) 2.7 cm** |

| the perpendicular bisector | | |
|---|---|---|
| **2.9** |  | **Equal arcs from $P$ and $Q$ above and below; the line crosses $PQ$ at 4.5 cm from each end at 90°** |
| **2.10** |  | **Equal arcs from $A$ and $B$ above and below; the line crosses $AB$ at 4 cm from each end at 90°** |
| **2.11** |  | **Equal arcs from $X$ and $Y$ above and below; the line crosses $XY$ at 5 cm from each end at 90°** |

| | | |
|---|---|---|
| **2.12** |  | **Equal arcs from $C$ and $D$ above and below; the line crosses $CD$ at 3 cm from each end at 90°** |
| **2.13** |  | **Equal arcs from $E$ and $F$ above and below; the line crosses $EF$ at 5.5 cm from each end at 90°** |
| **2.14** |  | **Equal arcs from $M$ and $N$ above and below; the line crosses $MN$ at 2.5 cm from each end at 90°** |